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Research Article

Designing a Scalable ELT Pipeline for Airbnb Analytics Using Medallion Architecture and Cloud-Based Data Engineering

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Abstract: The rapid growth of digital platforms has increased the need for scalable data infrastructures capable of processing large volumes of heterogeneous data for business analytics. In the short-term rental industry, Airbnb generates extensive operational data that can provide valuable business insights when integrated with external demographic information. However, many organizations still face challenges related to data silos, inconsistent schemas, limited traceability, and inefficient manual workflows. This study proposes an end-to-end Extract, Load, Transform (ELT) pipeline architecture that integrates Airbnb listing data in Sydney with demographic datasets from the Australian Bureau of Statistics using Google Cloud Storage, Apache Airflow, PostgreSQL, and dbt within a Medallion Architecture framework. The Bronze layer handled raw data ingestion, the Silver layer performed cleaning and historical snapshot transformations, and the Gold layer generated fact tables, dimension tables, and analytical data marts. The results demonstrate that the proposed architecture improves scalability, automation, data quality, and traceability while enabling advanced business analysis related to regional profitability, host distribution, property performance, and mortgage coverage. The study contributes to modern data engineering literature by demonstrating how integrated ELT architectures can support reproducible analytics for platform-based business environments.

Keywords: ELT Pipeline, Data Engineering, Airbnb Analytics, Medallion Architecture, Apache Airflow

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1. Introduction

The rapid growth of digital platforms has significantly increased the volume, variety, and velocity of data generated across industries, creating new challenges for organizations in managing scalable and reliable analytical infrastructures. In the short-term rental industry, platforms such as Airbnb continuously generate large-scale transactional and operational data, including listing attributes, pricing information, occupancy trends, host activities, and customer behavior patterns. However, organizations often struggle to integrate heterogeneous datasets originating from multiple sources such as CSV repositories, cloud storage systems, and external governmental databases due to issues related to data silos, inconsistent schema structures, poor traceability, and limited scalability. Recent studies have highlighted a growing transition from traditional ETL frameworks toward cloud-based ELT architectures that enable faster processing, higher flexibility, and improved scalability for modern analytics workloads (Yasmin et al., 2024; Astronomer, 2024). Furthermore, integrating operational platform data with external demographic datasets has become increasingly important for generating richer business insights, particularly in digital marketplace ecosystems and property analytics domains (Sserunjogi et al., 2025). Despite these developments, building an automated and scalable pipeline that effectively integrates

platform-generated rental data with regional demographic information remains a significant challenge.

To address these challenges, modern data engineering practices increasingly adopt workflow orchestration and transformation tools such as Apache Airflow and dbt to improve automation, reproducibility, and maintainability of data pipelines. Apache Airflow enables automated scheduling and monitoring of extraction and loading processes through Directed Acyclic Graphs (DAGs), improving operational efficiency and reducing manual intervention in large-scale data workflows (Kerner, 2024). Meanwhile, dbt supports modular SQL-based transformations that improve data quality, version control, and documentation across analytical pipelines. In parallel, the emergence of the Medallion Architecture framework—consisting of Bronze, Silver, and Gold layers—has become a widely adopted approach for structuring modern data warehouses by progressively improving data quality from raw ingestion to business-ready analytical outputs (Databricks, 2023). Recent research demonstrates that layered architectures significantly enhance pipeline maintainability, reduce redundancy, and improve analytical performance in big data environments (Picatto et al., 2024). Nevertheless, limited studies have explored the implementation of this architecture in short-term rental analytics while integrating demographic census data for broader socioeconomic analysis.

Based on this research gap, this study proposes an end-to-end cloud-based ELT pipeline that integrates Airbnb listing data in Sydney with demographic datasets from the Australian Bureau of Statistics using Google Cloud, Apache Airflow, PostgreSQL, and dbt within a Medallion Architecture framework. The Bronze layer handles raw data ingestion from cloud storage, the Silver layer performs cleaning, standardization, and historical snapshot processing, and the Gold layer develops fact tables, dimension tables, and analytical data marts for business intelligence purposes. This study contributes both theoretically and practically by demonstrating how scalable ELT pipelines can support reproducible analytics workflows while generating strategic insights related to regional profitability, host behavior, and property performance in the digital rental economy. Ultimately, this research extends the application of modern data engineering frameworks in real-world platform analytics and provides a scalable model for future data-driven decision-making systems.

2. Preliminaries

Recent literature has emphasized the growing importance of modern data engineering architectures in addressing the limitations of traditional ETL systems, particularly when handling high-volume, heterogeneous, and continuously evolving datasets. Traditional ETL pipelines often struggle with scalability, schema evolution, and real-time analytical requirements due to their rigid transformation processes prior to data loading. To overcome these limitations, researchers have increasingly shifted toward ELT architectures, where raw data is first stored and transformed later within scalable cloud environments. Apache Airflow has emerged as one of the most widely adopted orchestration platforms due to its ability to automate workflow scheduling, dependency management, and monitoring through Directed Acyclic Graphs (DAGs). Yasmin et al. (2024) found that workflow-as-code platforms significantly improve pipeline reproducibility but still face challenges related to debugging complexity and operational maintenance. Similarly, Mbata et al. (2024) highlighted that modern pipeline tools such as Airflow, Dagster, and Prefect improve automation efficiency while reducing manual operational overhead in large-scale analytical systems. Meanwhile, Bardsley et al. (2021) demonstrated that cloud-native pipeline architectures improve cost optimization and processing latency through distributed computing approaches, making them increasingly suitable for enterprise-scale analytics workloads.

The transformation layer has also received significant scholarly attention, particularly with the emergence of dbt and Medallion Architecture frameworks that improve data modularity and analytical readiness. Databricks introduced the Bronze-Silver-Gold layered architecture as a structured approach for improving data quality incrementally across analytical pipelines. Mohna et al. (2022) explained that Medallion Architecture enhances traceability, governance, and reproducibility by separating raw ingestion, cleaned transformation, and business-ready datasets into distinct layers. Likewise, Armbrust et al. (2022) introduced the lakehouse paradigm as a hybrid model that combines the flexibility of data lakes with the governance capabilities of traditional data warehouses. Zhang et al. (2023) further emphasized that metadata management and schema evolution remain critical factors in maintaining scalable cloud pipelines, particularly in environments with frequent structural

changes. Recent studies also show that modular SQL transformation frameworks such as dbt significantly improve maintainability and documentation across analytical workflows, reducing technical debt in long-term data engineering projects (Seenivasan, 2024).

Despite these advancements, limited studies have applied modern ELT architectures to platform-based property analytics that integrate operational marketplace data with external demographic datasets. Existing studies on Airbnb analytics largely focus on pricing prediction, recommendation systems, and consumer behavior analysis rather than data infrastructure development. Recent publications from Airbnb Engineering (2025) highlight the increasing role of large-scale data infrastructure in supporting marketplace intelligence, personalization, and forecasting systems. However, empirical studies that combine Airbnb operational data with demographic datasets from institutions such as the Australian Bureau of Statistics remain scarce. This creates a research gap in demonstrating how cloud-based ELT pipelines can support integrated socioeconomic analytics within short-term rental ecosystems. Therefore, this study extends prior literature by implementing an end-to-end ELT pipeline using Google Cloud, Apache Airflow, PostgreSQL, and dbt to enable scalable analytics for Airbnb and census data integration.

3. Materials and Method

3.1. Data Source

The data sources used in this study consist of both platform-generated operational data and external demographic reference data to support comprehensive analytical modeling. The primary dataset was obtained from Airbnb monthly listing exports, which contain detailed information related to property listings, host characteristics, pricing attributes, availability, accommodation capacity, room types, and estimated revenue indicators across multiple periods in Sydney. These monthly datasets enable temporal analysis by capturing changes in listing performance over time. To enrich the operational rental data with socioeconomic context, this study also incorporates demographic datasets from the Australian Bureau of Statistics, specifically Census G01 and G02 datasets that provide regional indicators such as median age, household size, rental costs, mortgage repayment estimates, and population characteristics at the Local Government Area (LGA) level. In addition, LGA code mapping datasets and suburb-to-LGA reference tables were utilized to establish relational consistency between Airbnb neighborhood-level data and census geographic boundaries.

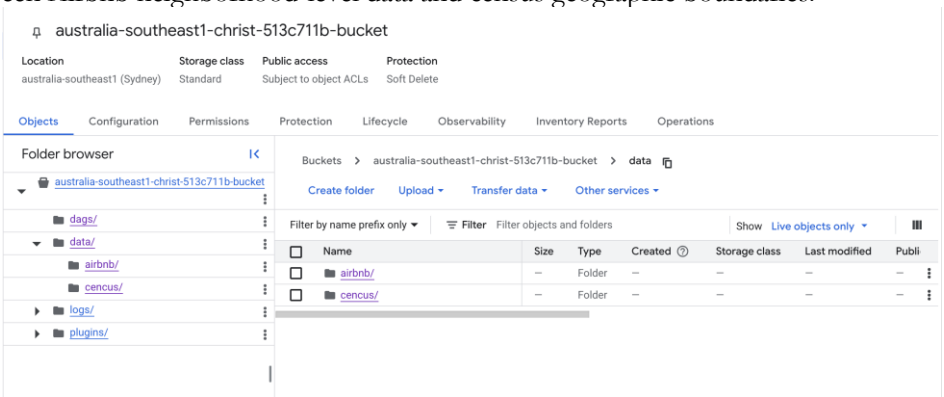


Figure 1. Google Cloud Storage Data Repository

Figure 1 shows the folder structure of the Google Cloud Storage (GCS) bucket used in this project (australia-southeast1-christ-513c711b-bucket). Under the root directory, the data/ folder contains two primary sub-folders: airbnb/ and cencus/. The airbnb/ folder stores monthly CSV exports of Airbnb listing data (e.g., 05_2020.csv, 06_2020.csv, etc.), while the cencus/ folder holds ABS Census 2016 datasets (G01, G02, LGA code, and LGA–suburb mapping files). This organization ensures a clear separation between business data and demographic reference data, supporting consistent ingestion paths for the Bronze Layer DAGs in Airflow.

3.2. Schema Design

The schema design in this study was developed to ensure data organization, traceability, and scalability throughout the ELT pipeline by implementing a structured Medallion

Architecture within PostgreSQL. In the Bronze layer, separate raw tables were created to preserve the original structure of each dataset, including Airbnb listings, Census G01 and G02 datasets, LGA code mappings, and suburb mapping tables. Each raw table retained all original attributes from the source files and included additional metadata fields such as source_file to track data lineage and load_timestamp to record ingestion time, which are considered essential practices for maintaining traceability in modern data systems (Zhang et al., 2023). In the Silver layer, staging tables were designed to perform data cleaning, schema standardization, type casting, null handling, and attribute normalization, while snapshot tables were implemented using dbt to capture historical changes through Slowly Changing Dimension (SCD Type 2) mechanisms. These snapshot schemas stored temporal attributes such as dbt_valid_from and dbt_valid_to to maintain historical consistency, aligning with best practices in temporal data warehousing (Kimball & Ross, 2021). In the Gold layer, a star-schema-oriented analytical model was developed consisting of fact tables, dimension tables, and data marts optimized for business intelligence and analytical querying. The fact table stored transactional metrics such as estimated revenue, occupancy, and listing activity, while dimension tables represented entities such as hosts, listings, suburbs, and Local Government Areas (LGAs). Additional analytical data marts were created to support business intelligence queries related to property performance, host distribution, and neighborhood-level profitability. Layered schema architectures such as Medallion have been shown to improve governance, reduce redundancy, and enhance analytical performance in cloud-based environments (Armbrust et al., 2022; Mohna et al., 2022). Therefore, this schema design ensures that downstream analytical processes operate on clean, reliable, and business-ready datasets.

3.3. Automation and Orchestration

Automation and orchestration play a critical role in ensuring the reliability, repeatability, and efficiency of modern ELT pipelines, particularly when handling large-scale and continuously evolving datasets. In this study, Apache Airflow was implemented as the primary workflow orchestration platform to automate data ingestion processes from Google Cloud Storage into PostgreSQL. A Directed Acyclic Graph (DAG) named bronze_load_first_time was developed to manage the initial ingestion of all raw datasets, including Airbnb listings, census datasets, and geographic mapping files. Each task within the DAG was responsible for extracting CSV files from cloud storage, performing preliminary preprocessing tasks such as standardizing column names, removing redundant attributes, and validating schema compatibility before loading the data into the Bronze layer tables. Workflow orchestration platforms such as Airflow have been widely recognized for improving scheduling efficiency, dependency management, fault tolerance, and workflow reproducibility in data engineering environments (Yasmin et al., 2024). Additionally, studies by Verma et al. (2023) found that automated orchestration frameworks significantly reduce operational overhead and improve pipeline reliability in cloud-based infrastructures. Airflow's integrated monitoring dashboard also enables administrators to track task execution, identify failures, and rerun processes when necessary, improving operational visibility and maintenance efficiency (Astronomer, 2024). Furthermore, automated orchestration has been shown to enhance scalability and reduce human error in complex analytical pipelines that require continuous incremental updates (Mbata et al., 2024). Therefore, implementing automated orchestration mechanisms ensures consistent workflow execution while minimizing manual intervention across the proposed data pipeline architecture.

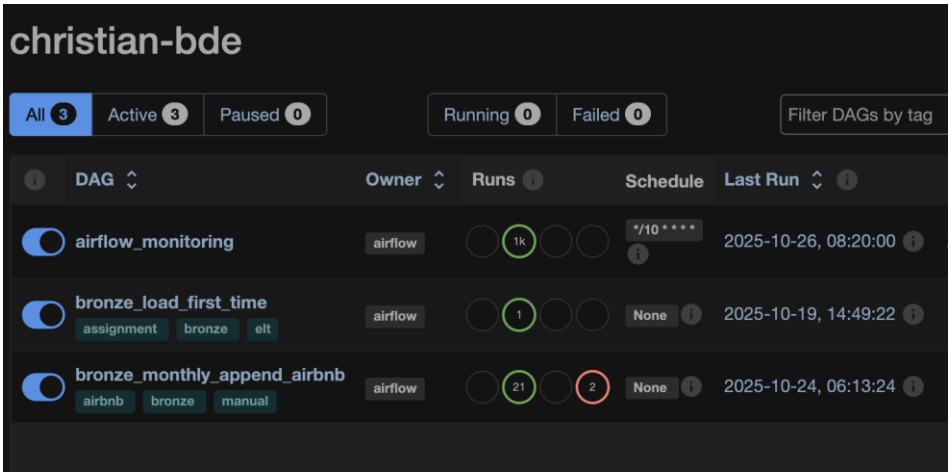


Figure 2. Building ELT data pipelines with Airflow.

3.4. Monthly Incremental Loads

The To support continuous updates, a secondary DAG bronze_monthly_append was designed.

- This workflow enables manual, month-by-month appending of new Airbnb CSV files while preserving historical records.
- Users can manually trigger each run with a JSON configuration specifying the month (e.g., {"execution_month": "06_2020"}).
- This approach ensures temporal consistency and maintains data integrity across sequential loads

3.5. Quality and Reliability Measures

Quality and reliability measures were implemented to ensure data integrity, consistency, and operational stability throughout the ELT pipeline. Since the pipeline integrates heterogeneous datasets from Airbnb and the Australian Bureau of Statistics, several validation mechanisms were introduced to minimize errors during ingestion and transformation processes. At the Bronze layer, schema validation checks were applied to ensure column consistency and prevent schema drift when new monthly datasets were appended. Automated preprocessing routines removed duplicate records, eliminated unnecessary attributes such as unnamed columns, and standardized naming conventions to maintain structural consistency. Logging mechanisms were also implemented to record row counts, ingestion timestamps, and source file paths, enabling end-to-end data lineage tracking and auditability. In the Silver layer, additional quality controls included null-value handling, datatype validation, referential integrity checks between suburb and Local Government Area (LGA) mapping tables, and snapshot testing using dbt to ensure historical consistency. Recent studies emphasize that automated data validation frameworks significantly improve pipeline reliability by detecting anomalies early and reducing downstream analytical errors (Zhang et al., 2023). Similarly, Monte Carlo (2024) reported that proactive data observability practices such as anomaly detection, schema monitoring, and lineage tracking improve trustworthiness in large-scale analytical systems. Furthermore, research by Rekatsinas et al. (2022) highlighted that implementing automated data quality checks can substantially reduce the risk of inaccurate business insights caused by incomplete or inconsistent datasets. These quality assurance mechanisms ensure that the proposed pipeline remains robust, scalable, and reliable for long-term analytical use.

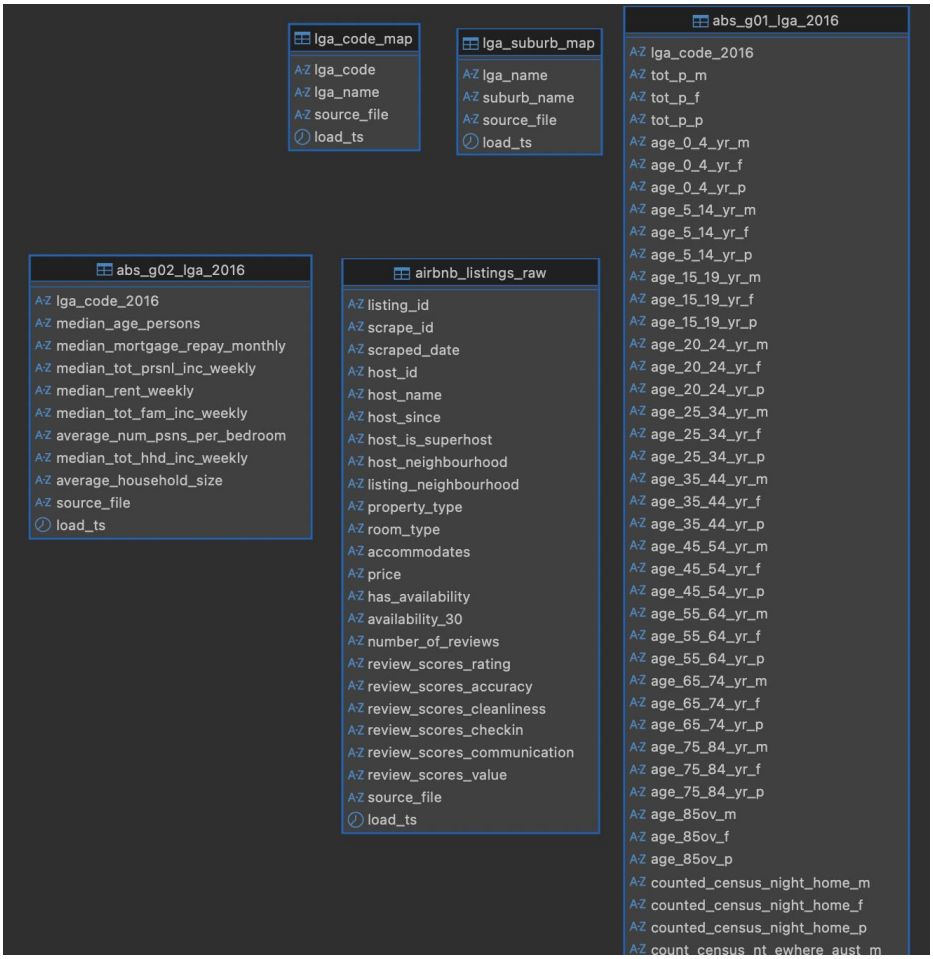


Figure 3. Bronze schema structure

Figure 3 shows the Bronze schema designed for raw data persistence and schema standardization. Each source—Airbnb listings, ABS demographic datasets (G01 and G02), and LGA mapping tables—is stored in separate, schema-aligned tables to preserve data granularity and lineage. This modular structure isolates ingestion domains, simplifies incremental loads, and ensures referential integrity between Airbnb neighbourhoods and LGA boundaries through mapping relationships. Fields like source_file and load_ts enable traceable version control for downstream Silver transformations.

3.6. Monthly Incremental Loads

At the completion of this stage:

- All raw CSV files were successfully ingested into the PostgreSQL Bronze schema.
- The pipeline now supports incremental monthly updates via Airflow triggers.
- The data is preserved in its original form, ready for transformation in the Silver Layer (staging and snapshot models).

This marks the completion of the data-landing phase and provides a stable starting point for downstream modeling and analysis

4. Results

4.1. Silver Layer

Figure 4 illustrates the Silver layer, which bridges the raw Bronze data and the analytical Gold layer. This layer consists of two main stages: data staging and dbt snapshotting. Each staging model (prefixed stg_) standardizes column names, types, and values derived from the raw ingestion tables, while snapshot models (prefixed snap_dim_) capture temporal changes for dimension tracking.

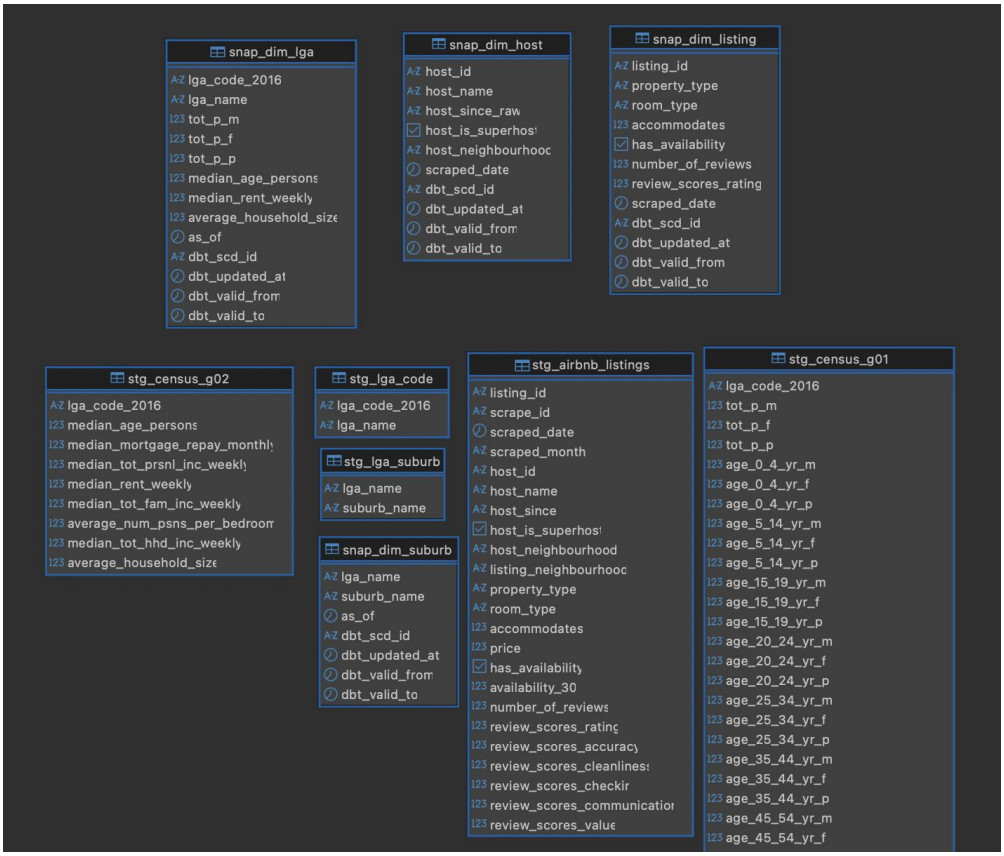


Figure 4. Silver Layer: Staging and Snapshot Architecture

During the cleaning stage, several key transformations are performed:

- Data type casting: numeric and date fields such as price, accommodates, and scraped_date are explicitly cast to maintain analytical consistency.
- Whitespace and null normalization: empty strings are replaced with NULL, and inconsistent capitalization or spacing is removed for attributes like property_type and room_type.
- Boolean standardization: textual flags such as "t" and "true" are normalized into proper Boolean values.
- Date alignment: snapshot timestamps (scraped_date, as_of) are standardized into ISO date formats to ensure version tracking.
- Referential consistency: joins between LGA, suburb, and listing datasets are verified through unified key fields (e.g., lga_code_2016 and lga_name).

The snapshot stage then applies dbt's timestamp strategy to detect and record data changes over time. Each snapshot—snap_dim_listing, snap_dim_host, snap_dim_lga, and snap_dim_suburb—maintains columns like dbt_valid_from and dbt_valid_to, providing a built-in SCD Type 2 mechanism for tracking entity evolution. This architecture ensures that every downstream model in the Gold layer references a clean, versioned, and historically accurate dataset, enabling point-in-time analysis across multiple reporting periods.

4.2. Golden Layer

This layered structure enables progressive refinement of data quality and business meaning, ensuring that each subsequent stage only depends on verified, versioned outputs. The approach aligns with best practices in modern data engineering, where each layer enforces data lineage, modularity, and analytical reproducibility.

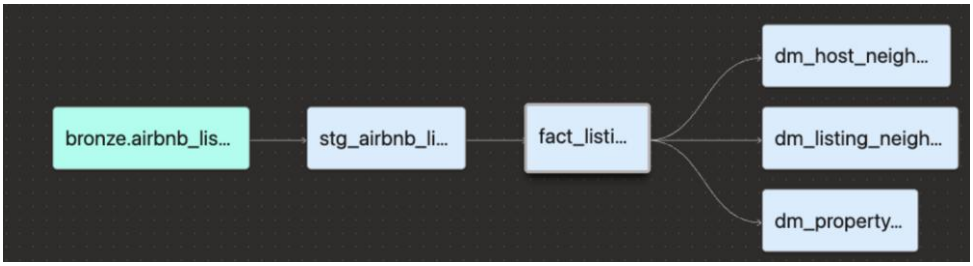


Figure 5. Create a data warehouse architecture on Postgres using the Medallion architecture (Bronze, Silver, Gold) with dbt. Include at least 4 dimension tables (e.g., host, suburb, LGA, etc.) along with two Census tables as reference data in the Gold layer.

This diagram in Figure 5, illustrates the complete dbt-based data warehouse pipeline built on Postgres following the Medallion architecture

- The Bronze layer stores raw ingested Airbnb and Census datasets directly from Google Cloud Storage.
- The Silver layer (e.g., stg_airbnb_listings, snap_dim_host, snap_dim_lga, snap_dim_suburb) performs data cleaning, type casting, deduplication, and snapshot versioning, ensuring historical traceability for every entity.
- The Gold layer contains fact and dimension tables optimized for analytics:
  - Fact table: fact_listing, aggregating monthly revenue, availability, and stay metrics.
  - Dimension tables: dim_host, dim_listing, dim_suburb, dim_lga, each sourced from corresponding snapshots in Silver.
  - Reference tables: Census datasets (abs_g01_lga_2016 and abs_g02_lga_2016) provide demographic and economic context for enriched analysis.
  - Mart models (dm_host_neighbourhood, dm_listing_neighbourhood, dm_property_summary) compute higher-level insights on spatial, host, and property performance.

4.3. DataMart

4.3.1. Property Type Analysis (dm_property_type)

This view aggregates metrics by property characteristics. The lineage starts from stg_airbnb_listings, which feeds both snap_dim_listing and snap_dim_host snapshots. These snapshots then materialized into dim_listing and fact_listing, which collectively support the dm_property_type model. This mart summarizes key KPIs such as revenue, occupancy, and rating distribution across property categories.

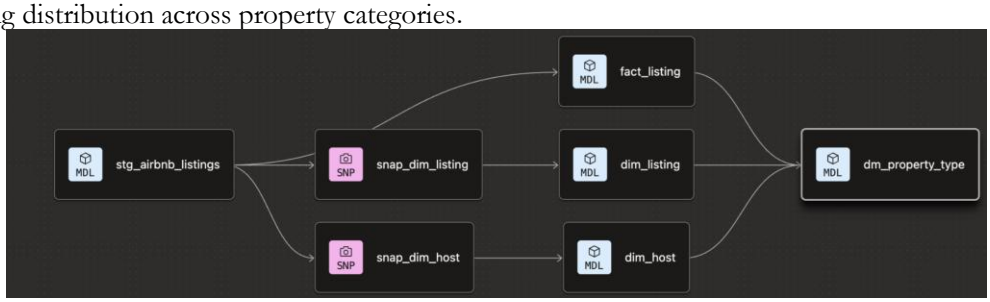


Figure 6. Property Type View Lineage

4.3.2. Host-Neighbourhood Analysis (dm_host_neighbourhood)

The second lineage highlights the flow toward dm_host_neighbourhood. The process begins with stg_airbnb_listings and its corresponding host snapshot (snap_dim_host), which is refined into the dim_host dimension. Joined with fact_listing, the mart computes host-level metrics like the number of listings, average stay rate, and total estimated revenue per neighbourhood.

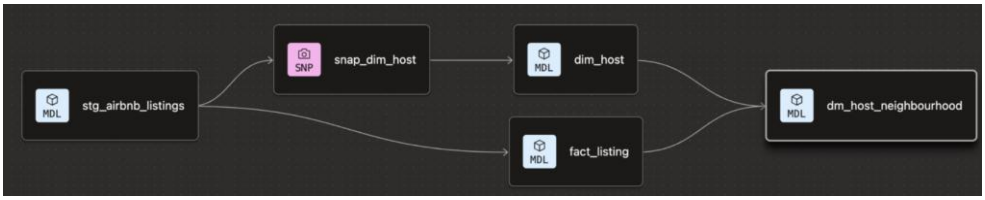


Figure 7. Host Neighbourhood View Lineage

4.3.3. Listing-Neighbourhood Analysis (dm_listing_neighbourhood)

The final lineage demonstrates a combined analytical perspective integrating both snap_dim_host and snap_dim_listing. From these snapshots, dim_host and dim_listing are built, then merged with fact_listing to construct dm_listing_neighbourhood. This model enables granular insight into neighbourhood performance — linking listing attributes, host behaviour, and geographical patterns for comparative analytics.

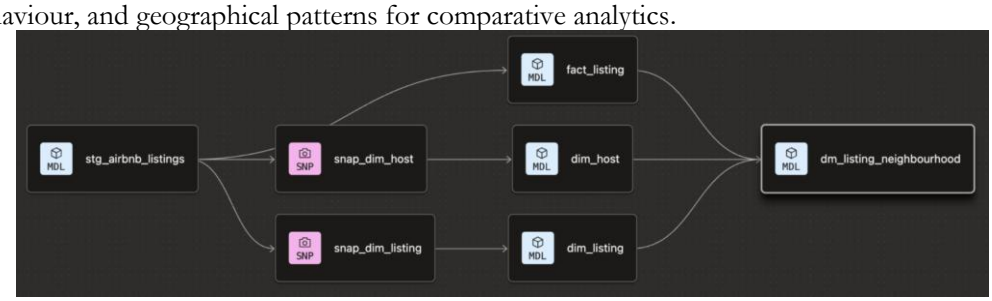


Figure 8. Listing Neighbourhood View Lineage

4.4. Airflow-Driven Data Ingestion

The data pipeline's operational workflow is orchestrated using Apache Airflow, which automates the ingestion of both Airbnb and Census datasets from Google Cloud Storage (GCS) into the Bronze schema on Postgres.

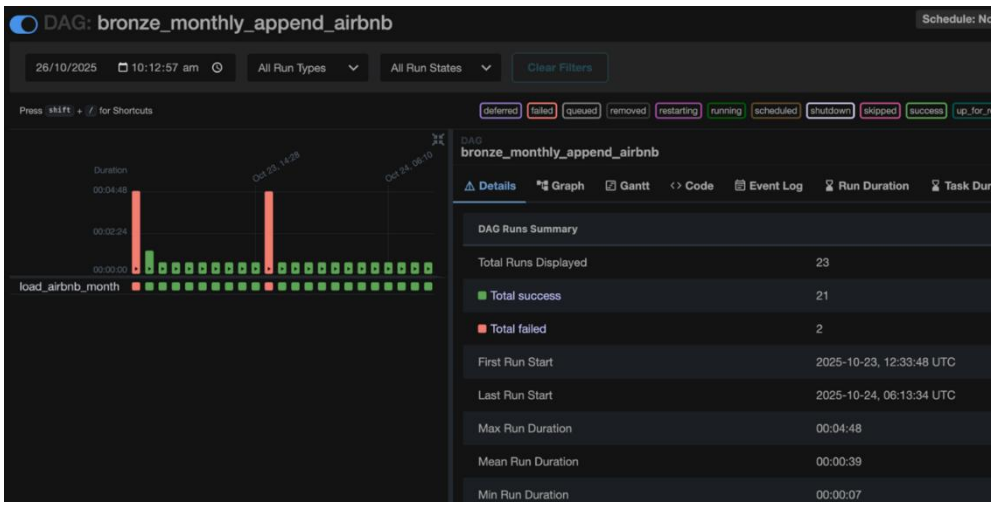


Figure 9. Airflow DAG Monitoring for Monthly Airbnb Data Appends

Two DAGs were implemented:

- bronze_load_first_time — performs the initial one-time ingestion of all static datasets (Airbnb May 2020 and Census tables).

- bronze_monthly_append_airbnb — supports incremental monthly appends of Airbnb data, ensuring chronological integrity and traceability.

Each DAG task reads CSV files directly from /home/airflow/gcs/data/..., cleans redundant columns, and appends records to the corresponding Bronze tables via SQLAlchemy using Airflow's PostgresHook. This modular design allows each dataset to be re-run or extended independently without re-creating historical data.

4.5. Sequential Monthly Loading

The monthly append DAG is configured to execute manually through Airflow's Trigger DAG interface, where users supply a JSON configuration, for example:

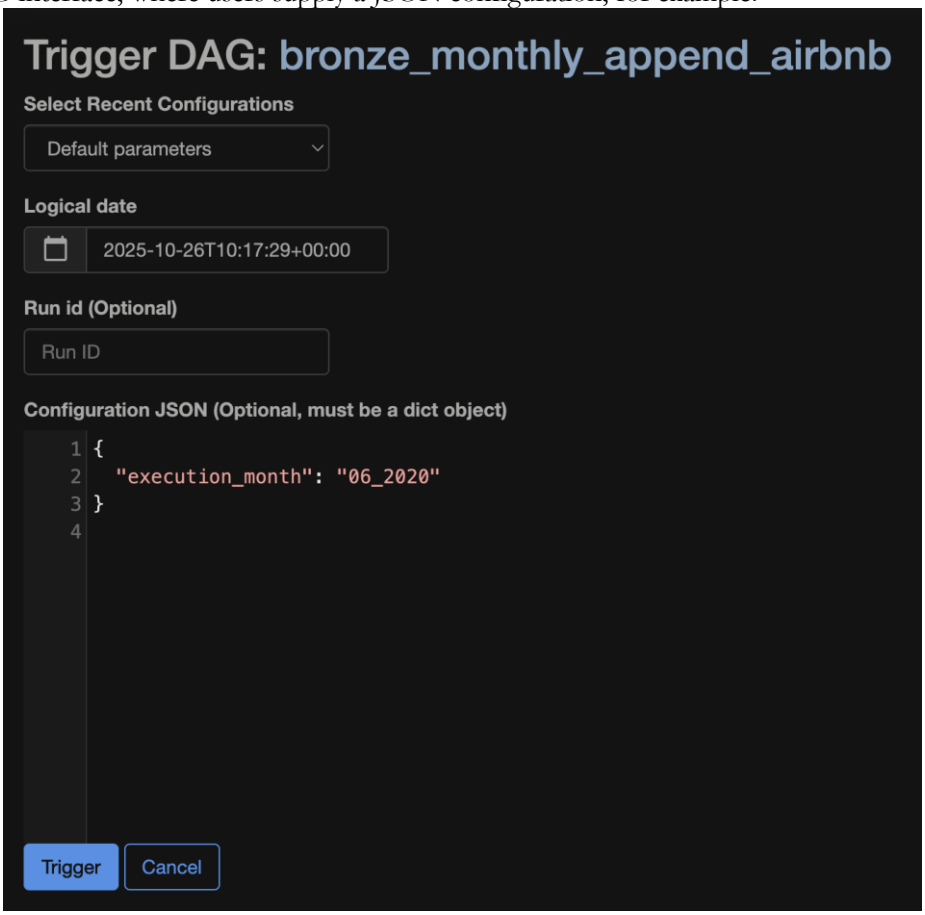


Figure 10. Manual trigger with config on Airflow.

This ensures every dataset is loaded sequentially in temporal order (May → June → July → ...) to preserve consistency in later time-based aggregations. Once each month's data is appended to the Bronze schema, it is available for transformation in dbt's Silver and Gold layers.

4.6. Ad-hoc Analysis

4.6.1. Demographic Differences Between Top and Bottom Performing LGAs

The comparison shows a clear socio-economic divide. The top-performing LGAs are typically affluent coastal and inner-city suburbs with older, smaller households and higher rental prices—factors that align with higher Airbnb profitability. In contrast, the lower-performing LGAs are characterized by younger populations, larger household sizes, and lower rents, suggesting stronger residential density but lower short-term rental demand. These

patterns emphasize that Airbnb market potential is shaped not only by property availability but also by the demographic and economic profiles of local communities.

Table 1. Demographic Comparison Between High-Performing and Low-Performing LGAs

LGA Name	Performance Group	Average Estimated Revenue (AUD)	Median Age (Years)	Average Household Size	Median Rent per Week (AUD)
Northern Beaches	Top 3 Performing	15,566.05	40	2.7	565
Woollahra	Top 3 Performing	15,313.22	39	2.3	650
Waverley	Top 3 Performing	11,149.98	35	2.4	622
Parramatta	Bottom 3 Performing	1,117.81	34	2.8	440
Burwood	Bottom 3 Performing	969.80	33	2.9	500
Cumberland	Bottom 3 Performing	699.05	32	3.2	400

4.6.2. Correlation Between Median Age and Estimated Revenue

To examine whether demographic maturity influences Airbnb performance, a correlation analysis was conducted between median age (from Census 2016) and the average estimated revenue per active listing across all Sydney LGAs. The resulting Pearson correlation coefficient of 0.167 indicates a weak positive relationship between the two variables.

This finding suggests that areas with slightly older populations tend to generate marginally higher Airbnb revenues, but the relationship is not strong enough to be deterministic. In practice, while age composition may reflect socio-economic stability and housing value, other factors—such as property density, proximity to tourist destinations, and pricing strategy—likely play a larger role in driving listing profitability.

4.6.3. Best Listing Type for Top Performing Neighbourhoods

To identify the optimal property characteristics driving the highest occupancy and revenue, the analysis examined the top-five combinations of property type, room type, and guest capacity across Sydney's leading neighbourhoods. Metrics such as the average number of stays per month and the average estimated revenue were aggregated from the Gold layer.

Table 2. Correlation Analysis Between Median Age and Airbnb Estimated Revenue Across LGAs

Property Type	Room Type	Accommodates	Active Listings	Average Stays	Average Estimated Revenue (AUD)
Private room in apartment	Private room	4	12	30.00	127 500.00
Apartment	Private room	4	12	30.00	127 500.00
Entire apartment	Entire home/apt	3	6	29.17	3 208.33
Townhouse	Entire home/apt	8	34	29.00	10 354.03
Apartment	Entire home/apt	8	11	28.73	6 931.82

The results reveal that private rooms within apartments consistently yield the highest estimated revenues, demonstrating exceptional occupancy (full 30 days) and strong profitability despite moderate capacity (four guests). These listings combine affordability with urban accessibility, aligning well with sustained traveller demand.

By contrast, entire homes or townhouses, although more spacious, show slightly lower occupancy rates and revenue per listing, likely reflecting higher nightly prices and a narrower customer segment. The pattern suggests that maximizing occupancy efficiency, rather than property size, is the key to driving revenue in Sydney's Airbnb market—particularly in densely populated, high-demand neighbourhoods.

4.6.4. Correlation Between Median Age and Estimated Revenue

This analysis explores whether Airbnb hosts with multiple listings tend to concentrate their properties within a single Local Government Area (LGA) or distribute them across multiple regions. Understanding this pattern provides insight into host business strategies and market focus.

Table 3. Top Property Type Combinations in High-Performing Neighborhoods

Host Type	Number of Hosts	Percentage (%)
Single-LGA Host	17,313	99.47
Multi-LGA Host	92	0.53

The results show a highly localized hosting pattern, with over 99% of hosts operating exclusively within a single LGA. This indicates that most Airbnb participants are small-scale or individual hosts who manage properties in their immediate area, rather than diversified commercial operators.

The 0.53% of multi-LGA hosts, although minimal, likely represent professional or institutional property managers who strategically expand into multiple high-demand zones. This sharp contrast underscores Airbnb's dominance by community-based hosts, emphasizing localized market knowledge and neighborhood familiarity as key success factors in Sydney's short-term rental ecosystem.

4.6.5. Correlation Between Median Age and Estimated Revenue

This analysis investigates whether the estimated annual Airbnb revenue per active listing is sufficient to cover the annualized median mortgage repayment in each LGA. The aim is to understand the financial viability of short-term rentals as a means of offsetting property ownership costs.

Table 4. Distribution of Single-LGA and Multi-LGA Hosts

LGA Name	Active Listings	Revenue-Covering Listings	Coverage Rate (%)	Avg. Estimated Revenue (AUD)
Northern Beaches	1,205	873	72.45	991,191.34
Mosman	228	156	68.42	910,316.62
Bathurst Regional	6	4	66.67	3,353,260.00
Upper Hunter	101	65	64.36	574,697.01
Waverley	2,739	1,762	64.33	677,321.11
Randwick	1,582	911	57.59	543,427.48
Woollahra	518	297	57.34	613,959.72
North Sydney	645	369	57.21	462,037.62
Hunters Hill	63	36	57.14	462,306.74

The data show that Northern Beaches leads the ranking, with over 72% of active hosts generating enough revenue to cover annual mortgage costs, followed closely by Mosman and Bathurst Regional. These areas combine high nightly rates, strong occupancy, and sustained visitor demand—enabling superior revenue performance. In contrast, inner-city LGAs such as Sydney, Randwick, and North Sydney show moderate coverage levels (~57–58%), likely due to elevated property prices and mortgage values outpacing average listing revenue.

Overall, the results indicate that Airbnb hosting can be financially sustainable in high-demand coastal or lifestyle regions, where occupancy and pricing synergy offset high property costs. Meanwhile, for urban hosts, profitability is more dependent on optimizing occupancy and pricing strategies rather than relying solely on location-based advantages

6. Conclusion

This study successfully developed an end-to-end ELT pipeline architecture for integrating short-term rental operational data from Airbnb with demographic datasets from the Australian Bureau of Statistics to support scalable business analytics. By leveraging Google Cloud Storage, Apache Airflow, PostgreSQL, and dbt within a Medallion Architecture framework, the study demonstrated how raw datasets can be systematically transformed into reliable, business-ready analytical assets through Bronze, Silver, and Gold layers. The implementation improved data traceability, automation, and scalability while enabling historical tracking through snapshot mechanisms and structured dimensional modeling. Additionally, the ad-hoc analytical results revealed that Airbnb profitability is influenced not only by property characteristics but also by broader demographic factors such as household composition, rental costs, and regional socioeconomic conditions.

From an analytical perspective, the developed data marts enabled deeper exploration of host behavior, neighborhood performance, and financial sustainability across different Local Government Areas (LGAs) in Sydney. The findings showed that high-performing regions tend to exhibit stronger socioeconomic indicators and that private-room apartment listings often generate higher occupancy and revenue efficiency compared to larger property types. Furthermore, most hosts were found to operate within a single LGA, indicating highly localized market behavior. Despite infrastructure limitations such as storage constraints and long transformation runtimes, optimization strategies such as incremental processing and query refinement improved overall system performance. This study contributes to the growing literature on modern data engineering by demonstrating a practical implementation of scalable ELT architecture for platform analytics and provides a foundation for future research involving real-time streaming pipelines, machine learning integration, and broader geographic datasets.

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Data Availability Statement: The datasets analyzed in this study consist of Airbnb monthly listing data and demographic datasets from the Australian Bureau of Statistics (ABS), including Census G01 and G02 datasets, LGA code mappings, and suburb-to-LGA reference data. The data supporting the findings of this study are available from the corresponding author upon reasonable request, subject to the terms and conditions of the respective data providers. The processed datasets generated during the study may also be made available by the corresponding author upon reasonable request.

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References

Armbrust, M., Karau, H., Xin, R., & Zaharia, M. (2022). Lakehouse: A new generation of open platforms that unify data warehousing and advanced analytics. Communications of the ACM, 65(9), 56–65.

Astronomer. (2024). *The state of Apache Airflow report 2024*. Astronomer.

Bardsley, D., Lu, L., & Shah, D. (2021). Serverless data pipeline design: Optimization under cost and latency constraints. Proceedings of the VLDB Endowment, 14(12), 3104–3117.

Databricks. (2023). *What is medallion architecture?* Databricks.

Kerner, S. M. (2024). Apache Airflow 2.10 arrives to usher in a new era of AI data orchestration. *VentureBeat*.

Mbata, A., Sripada, Y., & Zhong, M. (2024). A survey of pipeline tools for data engineering. arXiv preprint arXiv:2406.08335.

Mohna, H. A., Barua, T., Mohiuddin, M., & Rahman, M. M. (2022). AI-ready data engineering pipelines: A review of medallion architecture and cloud-based integration models. American Journal of Scholarly Research and Innovation, 1(1), 319–350.

Monte Carlo. (2024). The state of data reliability trends report. Monte Carlo Data.

Picatto, H., Heiler, G., & Klimek, P. (2024). Cost-effective big data orchestration using Dagster: A multi-platform approach. *arXiv preprint arXiv*.

Rekatsinas, T., Chu, X., Ilyas, I. F., & Ré, C. (2022). Holistic data quality management in modern data pipelines. Communications of the ACM, 65(1), 93–101.

Seenivasan, D. (2024). AI-driven enhancement of ETL workflows for scalable and efficient cloud data engineering. International Journal of Engineering and Computer Science, 13(6).

Sserunjogi, R., Ogenrwot, D., Niwamanya, N., et al. (2025). Design and evaluation of a scalable data pipeline for AI-driven monitoring systems in low-resource environments. *arXiv preprint arXiv*.

Verma, R., Singh, P., & Kumar, A. (2023). Cloud workflow orchestration for scalable data pipelines. Journal of Cloud Computing Systems, 11(2), 45–58

Yasmin, J., Wang, J., Tian, Y., & Adams, B. (2024). An empirical study of developers' challenges in implementing workflows as code: A case study on Apache Airflow. arXiv preprint arXiv.

Zhang, Q., Chen, L., & Zou, D. (2023). AI-driven metadata management and schema evolution in cloud data pipelines. ACM Transactions on Data Science, 4(2), Article 9.